



Bachelorarbeit

Algorithmische Ansätze für Globale Logistik im Co-Design

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1 Introduction

In today's globalized world, the production of commercial aircrafts has become a complex and highly distributed process. With production sites scattered across different countries, manufacturers rely on a vast network of internal facilities and suppliers. The production of an aircraft requires the transportation of large parts such as wings, engines, or fuselage sections between different production sites. To transport these large parts, special transportation means such as oversized road transportat, special aircraft freighters like the Beluga and special ships have to be used. The transportation of these large parts is not only costly, but also leads to high CO2 emissions. Therefore, finding a routing in the network minimizing logistics related costs and CO2 emissions is crucial. In this paper, we consider the problem of finding an optimal routing of products in a given global logistics network. The goal is to minimize the total cost of the network, while ensuring that the network is able to satisfy the demand for the products. We present a linear program formulation for this problem and solve it using Column Generation. Then we evaluate the performance of our algorithm on a set of test instances and present its limitations.

1.1 Problem Description

We are given a global logistics network $N(L, TL)$, where L is a set of Locations and $tl(\ell_i, \ell_j, R(tl)) \in TL$ are transport links from location ℓ_i to ℓ_j with $R(tl)$ being a set of routes associated with the transport link. A route $r(tr, distance) \in R(tl)$ has a transport resource $tr(cost, speed, co2emissions, capacity) \in TR$ and a $distance \geq 0$. The difference between a transport link and a route is that a transport link is an abstract connection between two locations and a route is a concrete connection between two locations using a specific transport resource. Through the network we want to send a set of given products $p(size, value, validTR) \in P$ from their production locations to their consumption locations. Each product p has a size, a value and a set of valid transport resources $validTR \subseteq TR$ which can be used to transport the product. Each product p can be produced at one or multiple locations at given fixed rates and consumed at one or multiple locations at given fixed rates. The goal is not only to find a routing of the products through the network, but also to find the corresponding transport resource packings, also named batches, for the routes. A transport resource packing is a set of products with their corresponding amounts which can be packed into a transport resource so that the total size of the products does not exceed the capacity of the transport resource. A solution should minimize the total cost of the network, while ensuring that the network is able to satisfy the demand for the products. The total cost of the network is the sum of the transportation costs, CO₂ emissions and the costs of tied-up capital. The transportation cost of a route is determined by the operating cost of the transport resource times the frequency of how often each transport resource packing is used to cover the demand for the products. To determine how many products can be transported on a route, we have to find one or multiple transport resource packings for each used route. Additionally the costs of CO₂ emissions are determined by how much CO₂ is emitted by the transport resource on the route times the frequency of how often each transport resource packing is used. Lastly the cost of tied-up capital is the opportunity cost of using the bounded capital of the products during transportation instead of using it for other potentially more profitable purposes. It is determined by the compound interest of the value of the products times the frequency of how often each transport resource packing is used. For each route r , we denote by B_r the set of transport resource packings for this route. A valid solution contains a set of transport resource packings $B \in B_r$ for every used route and their usage frequencies λ^B such that all constraints are satisfied and the total cost of the network is minimized. We summarize the parameters to reference them later on:

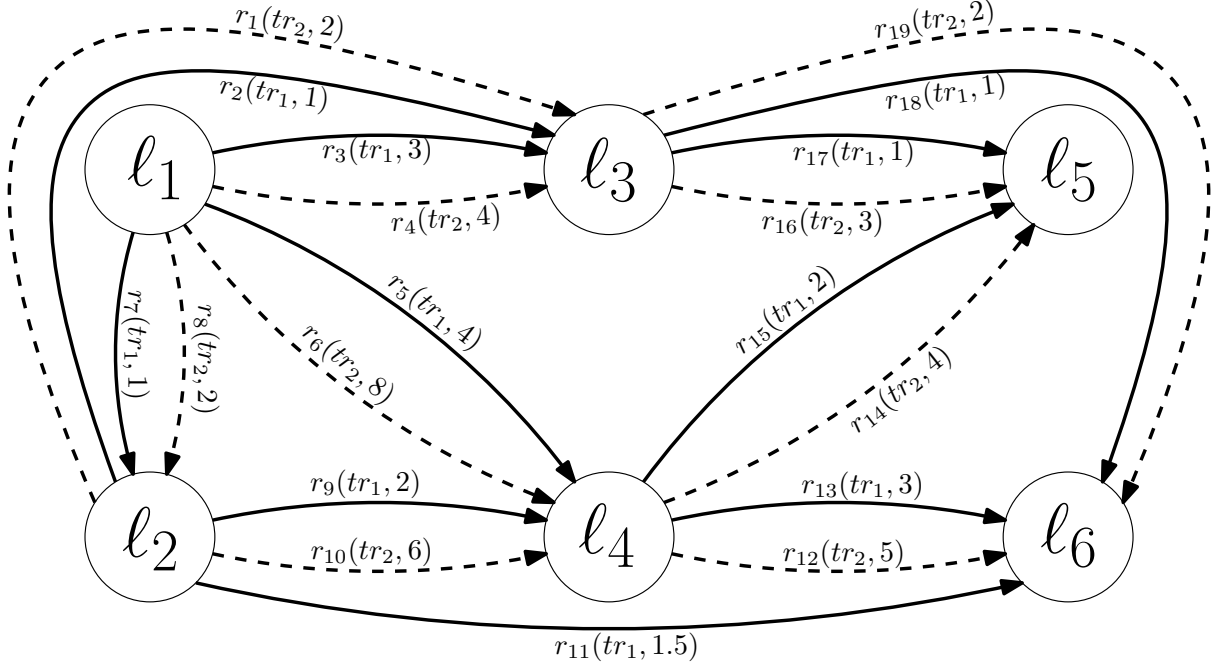


Figure 1: Example network

- $l \in L$: Location
- $r(tr, distance) \in R$: Route
- $tl(l_i, l_j, R(tr)) \in TL$: Transport Link
- $tr(cost, speed, co2emissions, capacity) \in TR$: Transport Resource
- $p(size, value, validTR) \in P$: Product
- B_r : Set of transport resource packings for route r
- λ^B : Usage frequency of transport resource packing $B \in B_r$

1.2 Example

As an introductory example we consider the following global logistics network $N(L, TL)$ with locations $L = \{l_1, l_2, l_3, l_4, l_5, l_6\}$ several unidirectional routes depicted in Figure 1, a CO₂ cost of 90 per ton and a yearly interest rate of 10%. Additionally the example includes two transport resources:

- $tr_1(cost, speed, co2emissions, capacity) = (56, 0.3, 60, 10)$ and
- $tr_2(cost, speed, co2emissions, capacity) = (42.5, 0.2, 45, 15)$.

All routes using transport resource tr_1 are visualized as solid lines and all routes using transport resource tr_2 are visualized as dashed lines.

At first we introduce one product $p_1(size, value, validTR) = (4, 1000, \{tr_1, tr_2\})$ to the network which is produced at l_1 at rate 15 and at l_2 at rate 20 and consumed at l_5 at a rate of 15

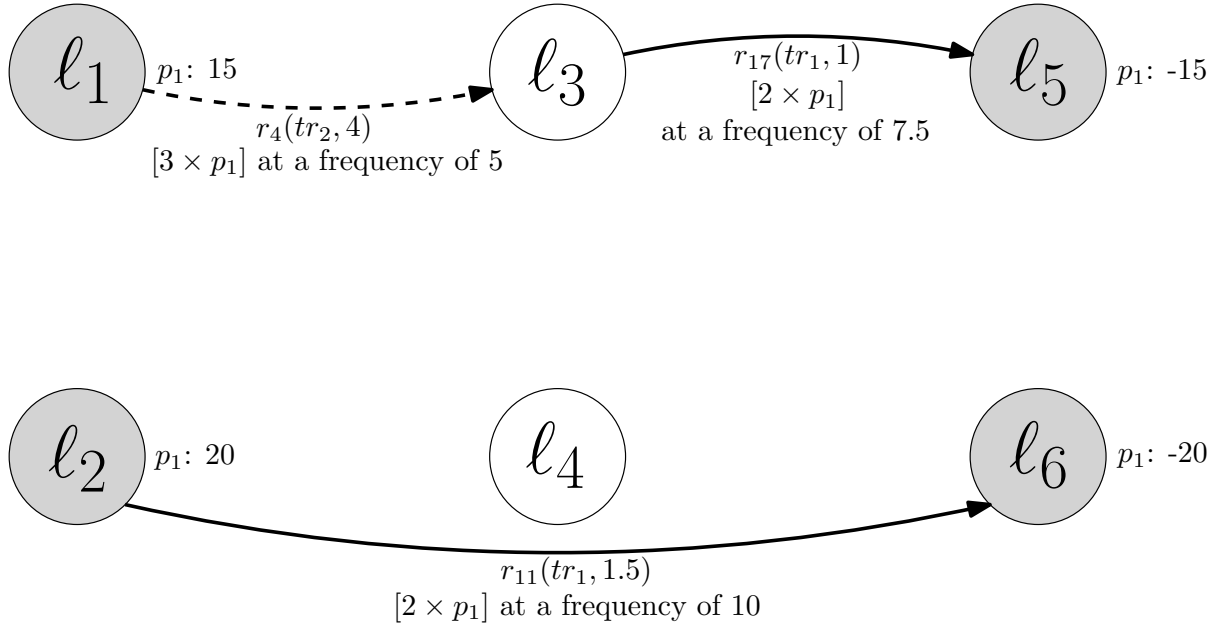


Figure 2: Example solution for product p_1

and at ℓ_6 at a rate of 20. If a product is consumed at a location it is visualized as a negative number in Figure 2 and Figure 3.

A solution for this problem instance is depicted in Figure 2 where the factories are colored in lightgrey and the intermediate locations are colored in white. The solution uses three routes r_4 , r_{11} and r_{17} each with corresponding transport resource packings and their usage frequencies:

- $r_1 : [3 \times p_1]$ at a frequency of 5,
- $r_{11} : [2 \times p_1]$ at a frequency of 10 and
- $r_{17} : [2 \times p_1]$ at a frequency of 7.5.

If we introduce a second product $p_2(size, value, validTR) = (3, 600, \{tr_1, tr_2\})$ to the network which is produced at ℓ_1 at a rate of 40 and consumed at ℓ_2 at a rate of 10, ℓ_5 at a rate of 20 and ℓ_6 at a rate of 10. A solution for this problem instance is depicted in Figure 3. The solution uses five routes r_2 , r_4 , r_8 , r_{11} and r_{17} each with corresponding transport resource packings and their usage frequencies:

- $r_2 : [1 \times p_1, 2 \times p_2]$ at a frequency of 9,
- $r_4 : [3 \times p_1, 1 \times p_2]$ at a frequency of 2,
- $r_8 : [5 \times p_2]$ at a frequency of 7 and $[3 \times p_1, 1 \times p_2]$ at a frequency of 3,
- $r_{11} : [1 \times p_1, 2 \times p_2]$ at a frequency of 5 and $[2 \times p_1]$ at a frequency of 7.5 and
- $r_{17} : [1 \times p_1, 2 \times p_2]$ at a frequency of 10 and $[2 \times p_1]$ at a frequency of 2.5.

We can observe that different products can be packed into the same transport resource packing and that the same product can be packed into different transport resource packings on the same route. Additionally we can observe that after we introduced the second product,

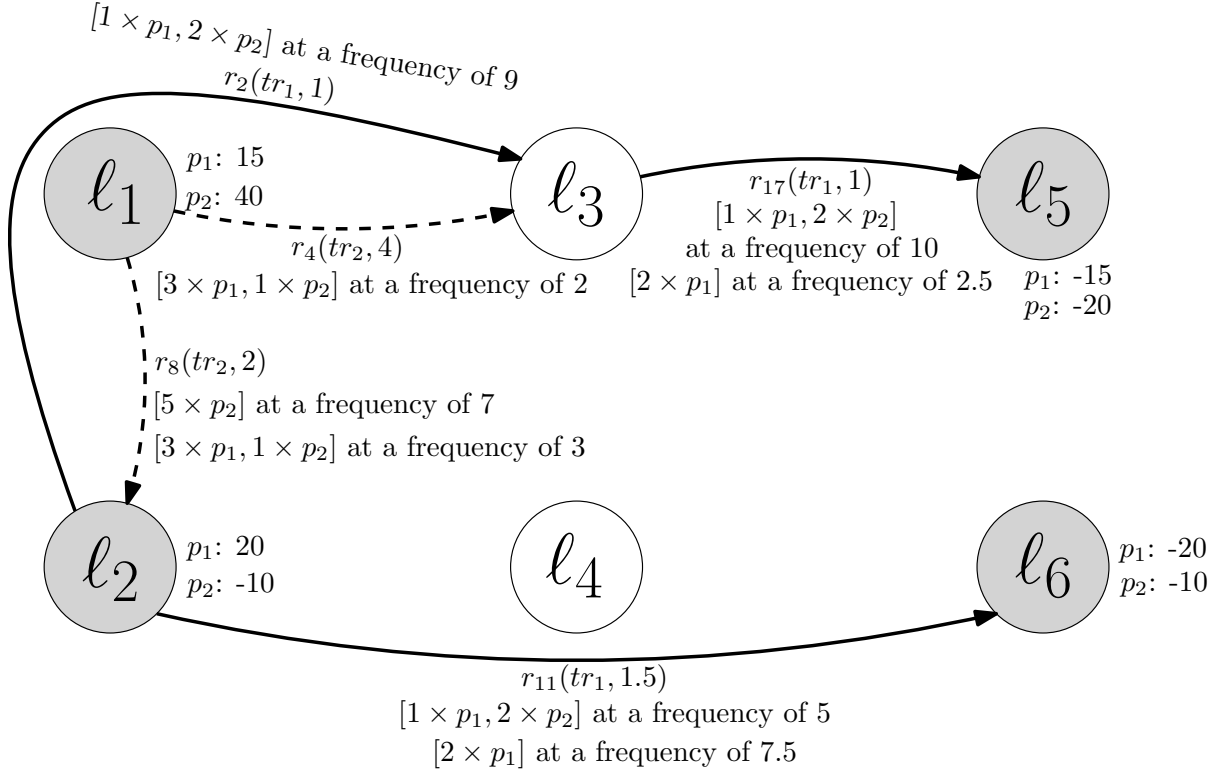


Figure 3: Example solution for product p_1 and p_2

the routing of the first product changed. In Figure 2 15 units of product p_1 are transported on route r_4 using transport resource packing $[3 \times p_1]$. In Figure 3 only 6 units of product p_1 are transported on route r_4 using transport resource packing $[3 \times p_1, 1 \times p_2]$ and 9 units of product p_1 are transported via route r_8 using transport resource packing $[3 \times p_1, 1 \times p_2]$ and route r_2 using transport resource packing $[1 \times p_1, 2 \times p_2]$. This is because we were able to find a new pattern which efficiently packs both products into the same transport resource packing on route r_8 : $[3 \times p_1, 1 \times p_2]$.

2 Background

2.1 Linear Programming

In this section we present the basics of linear programming referring to the book of R. Vanderbei [1, p. 6-8]. However, we only consider minimization problems. We start by introducing decision variables, the objective function and constraints. Then we present the standard form of a linear program and the concepts of feasible and optimal solutions. Finally we present examples of infeasible and unbounded problems.

Our Problem contains several variables which have to be decided in an optimal way e.g. how often a transport resource packing should be used. These variables are called decision variables. In linear programming, the decision variables are represented as a vector $\mathbf{x} = (x_1, \dots, x_n)^T$ where n is the number of decision variables. The decision variables are the variables whose values we aim to determine. The goal of a linear program is to find an optimal solution with respect to an objective function under some constraints. The objective function is a linear

function of the decision variables which we aim to minimize. It's of the form

$$\zeta = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

where $c_1, \dots, c_n$ are the coefficients of the objective function.

To restrict the solution space of the linear program, we introduce constraints. Constraints can be in the form of equalities or inequalities and are always linear combinations of the decision variables and the coefficients of the constraint. A constraint can be represented as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n \left\{ \begin{array}{l} \leq \\ \equiv \\ \geq \end{array} \right\} b$$

where $a_1, \dots, a_n$ are the coefficients of the constraint and b is the right hand side of the constraint. When we combine the objective function and the constraints, we get a linear program, which can be represented in the following standard form:

$$\begin{aligned} \min \quad & c_1x_1 + c_2x_2 + \cdots + c_nx_n \\ \text{subject to} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\ & \vdots \\ & a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m \\ & x_1, x_2, \dots, x_n \geq 0 \end{aligned} \tag{1}$$

where m is the number of constraints and n is the number of decision variables. A more compact representation of a linear program is given by

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \mathbf{c}^T \mathbf{x} \\ \text{subject to} \quad & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned} \tag{2}$$

where $\mathbf{c}$ is the vector of coefficients of the objective function, $\mathbf{x} = (x_1, \dots, x_n)^T$ is the vector of decision variables, $A \in \mathbb{R}^{m \times n}$ is the matrix of coefficients of the constraints and $\mathbf{b} = (b_1, \dots, b_m)^T$ is the vector of right hand sides of the constraints. The following pairwise equivalent equations show how to convert the constraints to their different forms:

$$A\mathbf{x} \leq \mathbf{b} \Leftrightarrow (-A)\mathbf{x} \geq -\mathbf{b} \tag{3}$$

$$A\mathbf{x} = \mathbf{b} \Leftrightarrow A\mathbf{x} \leq \mathbf{b} \text{ and } A\mathbf{x} \geq \mathbf{b} \tag{4}$$

$$A\mathbf{x} \leq \mathbf{b} \Leftrightarrow A\mathbf{x} + \mathbf{s} = \mathbf{b} \text{ and } \mathbf{s} \geq 0. \tag{5}$$

In equation (5) we introduced a new variable $\mathbf{s} = (s_1, \dots, s_m)^T$, called the slack variable, to each constraint to convert the inequalities to equalities.

To find a feasible solution for a linear program, we have to find values for the decision variables $\mathbf{x}$ which satisfies all constraints. Solutions which do not satisfy one or more constraints are called infeasible solutions. A solution is called optimal if it satisfies all constraints and minimizes the objective function. If the linear program has feasible solutions with arbitrarily large objective values, we call it unbounded.

According to duality theory, every linear program, also called primal linear program (P), has a corresponding dual linear program (D). The dual linear program (D) is given by

$$\begin{aligned} \text{(D)} \quad & \max_{\mathbf{y} \in \mathbb{R}^m} \mathbf{y}^T \mathbf{b} \\ \text{subject to} \quad & \mathbf{y}^T A \leq \mathbf{c}^T \\ & \mathbf{y} \geq 0 \end{aligned} \tag{6}$$

where $\mathbf{y}$ is a vector of decision variables for each constraint of the primal linear program.

2.2 Column Generation

Some problems have a huge number of variables J . In these cases it is computational expensive to store the whole program in memory and to solve it. If we have a look onto an optimal solution of e.g. the cutting stock problem, we can see that only very few variables are used in the optimal solution. All other variables are not needed to find the optimal solution. The idea of column generation is to start with a program which only contains a small and simple subset $J' \subseteq J$ of variables, also called the restricted master program (RMP). Because it is most likely that our RMP does not contain all variables which are required for an optimal solution to our original problem, an optimal solution to our RMP does not need to be an optimal solution to our original problem. To improve our RMP, we have to identify which variables to add to our RMP so that the objective value of our RMP decreases. This can be done by computing the reduced cost $\bar{c}_j = c_j - \sum_{i=1}^m a_{i,j} * y_i$ of each variable $j \in J \setminus J'$, where c_j is the coefficient of variable j in the objective function, y_i is the dual solution of our RMP of column i and $a_{i,j}$ is the column i of variable j . If the reduced cost of a variable $j \in J \setminus J'$ is negative, we can add it to our RMP. This process is called pricing. The pricing problem is the problem of finding a variable $j \in J \setminus J'$ with a negative reduced cost. According to M. Lübbecke [2, p. 1], Column generation acquires its finiteness and correctness properties from the simplex method. The process of column generation with explicit pricing is depicted in Algorithm 1.

Algorithm 1 Column Generation with explicit pricing

Input: restricted master program (RMP) with variables $J' \subseteq J$

Output: optimal solution to original problem

- 1: **repeat**
 - 2: Solve RMP with variables J' to optimality, obtain primal $\mathbf{x} \in \mathbb{R}^n$ and dual $\mathbf{y} \in \mathbb{R}^m$ solution
 - 3: Compute reduced costs $\bar{c}_j = c_j - \sum_{i=1}^m a_{i,j} * y_i$ for all $j \in J \setminus J'$ ▷ computationally prohibitive
 - 4: **if** there is a variable $j \in J \setminus J'$ with negative reduced cost $\bar{c}_j < 0$ **then**
 - 5: $J' \leftarrow J' \cup \{j\}$
 - 6: **end if**
 - 7: **until** there is no variable $j \in J \setminus J'$ with negative reduced cost $\bar{c}_j < 0$
-

Because enumerating all variables $j \in J \setminus J'$ and computing their reduced costs is computationally prohibitive, we solve a typically well structured auxiliary optimization problem called the pricing problem or the subproblem. The objective function of the pricing problem represents the reduced cost of the searched variable and the constraints have to restrict the solution space to all $j \in J$. It is important to note that one could also use a combinatorial algorithm to solve the pricing problem. Column generation with implicit pricing is depicted in Algorithm 2.

3 Related Work

3.1 Minimum Cost Multi Commodity Flow Problem

In the Minimum Cost Multi Commodity Flow Problem multiple commodities are shipped through the same network each having a source and a sink, where the directed edges have a cost per unit of flow and a capacity limitation, see S. Moradi et al. [3]. Our problem is strongly related to this problem, as we also have multiple commodities/products which are shipped through the same network. The difference is that we do not have capacity limitations

Algorithm 2 Column Generation with implicit pricing

Input: restricted master program (RMP) with variables $J' \subseteq J$

Output: optimal solution to original problem

```
1: repeat
2:   Solve RMP with variables  $J'$  to optimality, obtain primal  $\lambda^*$  and dual  $\pi^*$  solution
3:   solve  $z = \min\{\bar{c}_j \mid j \in J \setminus J'\}$  and obtain  $j^*$  ▷ pricing problem
4:   if  $z < 0$  then
5:      $J' \leftarrow J' \cup \{j^*\}$ 
6:   end if
7: until  $z \geq 0$ 
```

on the edges and that the cost of a route is not determined by the amount of flow on the route but by the transport costs, CO₂ emissions and the cost of tied up capital. Additionally in our problem commodities have multiple sources and sinks. To formally define the problem, we let $G = (N, A)$ be a graph with directed edges, where each edge (i, j) has a cost c_{ij} per unit of flow and a capacity limitation u_{ij} . Additionally there are k commodities which are produced each at a source $s_k \in N$ and consumed at a sink $t_k \in N$ at a given flow rate $flow_k$. The goal is to find a feasible flow $f_{i,j}^k$ for each edge $(i, j) \in A$ and each commodity $k \in K$ which minimizes the total cost of the network and satisfies the capacity limitations at each edge and the net supply or demand at each node.

The problem can be solved by using a linear program. We introduce a variable $f_{i,j}^k \geq 0$ for each edge $(i, j) \in A$ and each commodity $k \in K$, which represents the amount of commodity k transported on edge (i, j) . Besides the flow limitation constraint for each edge $(i, j) \in A$ and each commodity $k \in K$ we also have a flow conservation constraint for each node $n \in N$ and each commodity $k \in K$. The flow conservation constraint ensures that the amount of commodity k leaving a node n minus the amount of commodity k entering a node n corresponds to its production or consumption rate:

$$\min \sum_{(i,j) \in A} \sum_{k \in K} c_{ij} * f_{i,j}^k \quad (7)$$

subject to

$$\forall n \in N, \forall k \in K : \quad \sum_{(n,j) \in A} f_{n,j}^k - \sum_{(i,n) \in A} f_{i,n}^k = \begin{cases} flow_k & \text{if } n = s_k \\ -flow_k & \text{if } n = t_k \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

$$\forall (i, j) \in A, \forall k \in K : \quad f_{i,j}^k \leq u_{ij} \quad (9)$$

$$\forall (i, j) \in A, \forall k \in K : \quad f_{i,j}^k \geq 0 \quad (10)$$

This linear program is the basis for our linear program formulation of the problem described in Section 1.1.

3.2 Cutting Stock Problem

In the Cutting Stock Problem we have a set of n items of length $l_i \in \mathbb{Z}_+$ which have to be cut $d_i \in \mathbb{Z}_+, i \in [n] := \{1, \dots, n\}$ times from rolls of length of length $L \in \mathbb{Z}_+$. An optimal solution to the problem tries to minimize the number of used rolls and includes a set of used cutting patterns P while satisfying the demand of each item and ensuring that the total length of each cutting pattern is less or equal to the length of a roll. The article by P. C. Gilmore et al. [4]

presents a linear programming formulation for the problem. We introduce a variable λ_p for each cutting pattern $p \in P$, which represents the number of times the cutting pattern p is used. The goal is to minimize the number of used rolls, which is equal to the sum of all λ_p . The linear program is given by the following equation:

$$\min \sum_{p \in P} \lambda_p \quad (11)$$

subject to

$$\forall i \in [n] : \quad \sum_{p \in P} a_{p,i} * \lambda_p \geq d_i \quad (12)$$

$$\forall p \in P : \quad \lambda_p \in \mathbb{Z}_+ \quad (13)$$

where $a_{p,i}$ is the number of times item i is cut in cutting pattern p and the set of cutting patterns P is given by

$$P = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{Z}_+^n \mid \sum_{i=1}^n a_i * l_i \leq L \right\} \quad (14)$$

This problem can perfectly be mapped to our problem domain. If we replace:

- the cutting patterns P by the transport resource packings B_r for each route $r \in R(tl)$,
- the n items by the products P ,
- the demand d_i of each item by the flow demands $\mathbf{f}_{tl,p}$ of each transport link $tl \in TL$ for each product $p \in P$,
- the integer variable λ_p by the continuous variable λ^B for each transport resource packing $B \in B_r$ and each route $r \in R(tl)$,
- the length l_i of each item by the size of each product $p \in P$ and
- the length L of each roll by the capacity of each transport resource $tr(r)$

we get a linear program formulation which we can use as a basis for our problem:

$$\min \sum_{\substack{tl \in TL \\ r \in R(tl) \\ B \in B_r}} \lambda^B \quad (15)$$

subject to

$$\forall tl \in TL, \forall p \in P : \quad \sum_{\substack{r \in R(tl) \\ B \in B_r}} \alpha_p^B * \lambda^B \geq \mathbf{f}_{tl,p} \quad (16)$$

$$\forall tl \in TL, \forall r \in R(tl), \forall B \in B_r : \quad \lambda^B \geq 0 \quad (17)$$

where α_p^B denotes how often product p is packed into transport resource packing B and the set of transport resource packings B_r for each route $r \in R(tl)$ is given by

$$B_r = \left\{ \begin{pmatrix} \alpha_{p_1} \\ \vdots \\ \alpha_{p_n} \end{pmatrix} \in \mathbb{Z}_+^n \mid \sum_{i=1}^n \alpha_{p_i} * size(p_i) \leq capacity(tr(r)) \right\} \quad (18)$$

4 Modelling the problem

In this section we present a linear program formulation for the problem described in Section 1.1. We start by introducing the variables used in the formulation. Then we present the constraints and the objective function of the linear program and we prove that the linear program is equivalent to the problem described in Section 1.1. Finally we present a column generation approach to solve the linear program.

4.1 Master Problem

As there are many similarities between the problem described in Section 1.1 and the MINIMUM COST MULTI COMMODITY FLOW PROBLEM, we use a similar approach to model the problem. We introduce a variable $\mathbf{f}_{tl,p} \geq 0$ for each transport link $tl \in TL$ and each product $p \in P$. The variable $\mathbf{f}_{tl,p}$ represents the amount of product p transported on transport link tl .

For each location $\ell \in L$ and each product $p \in P$ we introduce a flow conservation constraint (19). The flow conservation constraint ensures that the amount of product p leaving a location ℓ minus the amount of product p entering a location ℓ is equal to the net supply or demand of product p at location ℓ . The net supply or demand of product p at location ℓ is given by the function $netSupplyDemand(\ell, p)$ which returns a positive rate if the product is produced at a location and a negative rate if the product is consumed at a location. The flow conservation constraint for each location $\ell \in L$ and each product $p \in P$ is given by the following equation:

$$\forall \ell \in L, \forall p \in P : \sum_{tl \in TL_{out}} \mathbf{f}_{tl,p} - \sum_{tl \in TL_{in}} \mathbf{f}_{tl,p} = netSupplyDemand(\ell, p) \quad (19)$$

where TL_{out} is the set of transport links going out of location ℓ and TL_{in} is the set of transport links going into location ℓ .

As we do not have a limitation on the number of transport resources used, we can transport an unlimited amount of products on each transport link $tl \in TL$. In contrast to the MINIMUM COST MULTI COMMODITY FLOW PROBLEM, we do not have fixed costs for each unit of flow. Instead we have fixed costs for each transport resource packing. To determine how many transport resource packings we need for each route $r \in R(tl)$, we introduce a variable $\lambda^B \geq 0$ for each transport resource packing $B \in B_r$, where B_r is the set of all transport resource packings on route r . The variable λ^B represents the number of times the transport resource packing B is used on a route r . It can be interpreted as the frequency of using transport resource $tr(r)$ with packing B on route r . Therefore it is a positive continuous variable. The number of transport resource packings needed for each route $r \in R(tl)$ is given by the following equation:

$$\forall tl \in TL, \forall p \in P : \sum_{\substack{r \in R(tl) \\ B \in B_r}} \alpha_p^B * \lambda^B \geq \mathbf{f}_{tl,p} \quad (20)$$

where α_p^B denotes how often product p is packed into transport resource packing B . The equation (20) ensures that the amount of product p transported on transport link tl covers the required flow amount given by variable $\mathbf{f}_{tl,p}$.

To determine the cost of a route, we introduce three different cost functions: the transportation cost, the CO₂ emission cost and the cost of tied-up capital. The transportation cost of a route is determined by how often transport resource packings we use on route r times the cost of the transport resource per distance unit $cost(tr(r))$ times the distance of the route $distance(r)$:

$$transportCost(r) = \sum_{B \in B_r} \lambda^B * cost(tr(r)) * distance(r) \quad (21)$$

The CO₂ emission cost of a route is determined by how often transport resource packings we use on route r times the CO₂ emissions of the transport resource in weight per distance unit times the distance of the route times the CO₂ cost per weight unit:

$$co2EmissionCost(r) = \sum_{B \in B_r} \lambda^B * co2emissions(tr(r)) * distance(r) * co2costs \quad (22)$$

To calculate the cost of tied-up capital, we introduce a function $compoundInterest(value, time)$ which returns the compound interest of a given value for a given time.

$$compoundInterest(value, time) = value * ((1 + interestRate)^{time} - 1) \quad (23)$$

The cost of tied-up capital is determined by how often transport resource packings we use on route r times the compound interest of the value of the products packed into the transport resource packing for the time it is tied-up on the route.

$$capitalCost(r) = \sum_{B \in B_r} \lambda^B * compoundInterest\left(\sum_{p \in P} \alpha_p^B * value(p), distance(r)/speed(tr(r))\right) \quad (24)$$

Therefore the total cost of the network is given by the following equation:

$$\min \sum_{\substack{tl \in TL \\ r \in R(tl)}} transportCost(r) + co2EmissionCost(r) + capitalCost(r) \quad (25)$$

This results in the following linear program formulation for the problem described in Section 1.1:

$$\begin{aligned} \min \quad & \sum_{\substack{tl \in TL \\ r \in R(tl)}} \sum_{B \in B_r} \lambda^B * cost(tr(r)) * distance(r) \\ & + \sum_{\substack{tl \in TL \\ r \in R(tl)}} \sum_{B \in B_r} \lambda^B * co2emissions(tr(r)) * distance(r) * co2costs \\ & + \sum_{\substack{tl \in TL \\ r \in R(tl)}} \sum_{B \in B_r} \lambda^B * \sum_{p \in P} \alpha_p^B * compoundInterest(value(p), distance(r)/speed(tr(r))) \end{aligned} \quad (26)$$

subject to

$$\forall l \in L, \forall p \in P : \quad \sum_{tl \in TL_{out}} f_{tl,p} - \sum_{tl \in TL_{in}} f_{tl,p} = netSupplyDemand(l, p) \quad (27)$$

$$\forall tl \in TL, \forall p \in P : \quad \sum_{\substack{r \in R(tl) \\ B \in B_r}} \alpha_p^B * \lambda^B \geq f_{tl,p} \quad (28)$$

$$\forall tl \in TL, \forall p \in P : \quad f_{tl,p} \geq 0 \quad (29)$$

$$\forall tl \in TL, \forall r \in R(tl), \forall B \in B_r : \quad \lambda^B \geq 0 \quad (30)$$

where TL_{out} is the set of transport links going out of location ℓ and TL_{in} is the set of transport links going into location ℓ . The set of transport resource packings B_r for each route $r \in R(tl)$ is given by

$$B_r = \left\{ \begin{pmatrix} \alpha_{p_1} \\ \vdots \\ \alpha_{p_n} \end{pmatrix} \in \mathbb{Z}_+^n \mid \sum_{p \in P} \alpha_p * size(p) \leq capacity(tr(r)) \right\} \quad (31)$$

4.2 Proof of equivalence

In this section we prove that the linear program formulation of section 4 is equivalent to the problem described in Section 1.1. We have to show that every feasible solution of the linear program formulation is also a feasible solution of the problem described in Section 1.1 and vice versa. Additionally, the objective value in both problems should be the same.

4.2.1 Proof of equivalence in feasible solution

A valid solution to the problem described in section 1.1 contains a set of transport resource packings $B \in B_r$ for every used route and their usage frequencies λ^B such that the demands are satisfied and the total cost of the network is minimized. Given a feasible solution to the problem described in Section 1.1, we know the set of transport resource packings $B \in B_r$ for every used route and their usage frequencies λ^B . This directly gives us the values for the variables λ^B in our linear program formulation and we know how often each product occurs in each transport resource packing α_p^B . Because the selection of the transport resource packing frequencies has to obey the demands in the real world problem, we know that the flow conservation constraint (27) is satisfied. Therefore, the constraint (28) is also satisfied. Because the flow and frequency values have to be positive in the real world problem, we know that the constraints (29) and (30) are also satisfied. Therefore, the solution is also feasible in our linear program formulation.

Given a feasible solution to our linear program formulation, we know the positive values of $\mathbf{f}_{tl,p}$ and that they satisfy the flow conservation constraint and the production and consumption rates at each location, see constraint (27). If in an optimal solution to our described problem a product gets produced at location $\ell \in L$ at rate 15 and consumed at the same location ℓ at rate 10, it is always most efficient to source the 10 units from the same location, leading to a net supply and demand of 5. Therefore, we know that all supply and demand rates of the linear program formulation are feasible solutions to the described problem. All generated transport resource packings $B \in B_r$ for each route $r \in R(tl)$ are also feasible solutions to the described problem, because they satisfy the capacity constraint of the transport resource $tr(r)$, see constraint (28). Because the transport packing frequencies have to cover the flow rates of the linear program formulation, we know that the constraint (28) is satisfied and therefore the solution to the linear program formulation is also feasible in the described problem.

4.2.2 Proof of equivalence in objective value

In our formulation the transportation costs, CO₂ emission costs and the cost of tied-up capital are directly based on the definitions given in the described problem. Therefore, the objective value of every feasible solution of our linear program formulation is equal to the objective value of every feasible solution of our described problem.

4.3 Restricted Master Problem

Because the size of the set of transport resource packings $|B_r|$ for each route r is huge, we use Column Generation to solve the linear program. We start with a small subset of transport resource packings $B'_r \subset B_r$ for each route r which also can be empty and iteratively add new transport resource packings to the set.

To ensure that the linear program is feasible in the first iteration, we use the Big-M method. By using this method we do not have to initialize the set of transport resource packings B'_r . We introduce a new helper variable $\mathbf{y}_{tl,p} \geq 0$ for each transport link $tl \in TL$ and each product $p \in P$ in constraint (34). The solver can choose to use this variable to satisfy the constraint

however, there is a penalty of M if the solver chooses to use this variable. The value of M is an upper bound for the cost of sending one unit of product p on transport link tl .

In each iteration we solve the Master Problem with the current set of transport resource packings B'_r for each route r and obtain an optimal primal solution and an optimal dual solution. Specifically we get a solution to the dual variables of constraint (34) called $\pi_{tl,p}$. Then we add a new transport resource packings $B \in B_r \setminus B'_r$ for every route r to the set B'_r if it reduces the objective value of the linear program. We stop the algorithm when no new transport resource packing can be added to these sets which reduces the objective value of the linear program.

$$\begin{aligned}
\min \quad & \sum_{\substack{tl \in TL \\ r \in R(tl)}} \sum_{B \in B'_r} \lambda^B * cost(tr(r)) * distance(r) \\
& + \sum_{\substack{tl \in TL \\ r \in R(tl)}} \sum_{B \in B'_r} \lambda^B * co2emissions(tr(r)) * distance(r) * co2costs \\
& + \sum_{\substack{tl \in TL \\ r \in R(tl)}} \sum_{B \in B'_r} \lambda^B * \sum_{p \in P} \alpha_p^B * compoundInterest(value(p), distance(r)/speed(tr(r))) \\
& + \sum_{\substack{tl \in TL \\ p \in P}} M * y_{tl,p}
\end{aligned} \tag{32}$$

subject to the following constraints:

$$\forall l \in L, \forall p \in P : \quad \sum_{tl \in TL_{out}} f_{tl,p} - \sum_{tl \in TL_{in}} f_{tl,p} = netSupplyDemand(l, p) \tag{33}$$

$$\forall tl \in TL, \forall p \in P : \quad \sum_{\substack{r \in R(tl) \\ B \in B'_r}} \alpha_p^B * \lambda^B + y_{tl,p} \geq f_{tl,p} \quad (\text{dual: } \pi_{tl,p}) \tag{34}$$

$$\forall tl \in TL, \forall p \in P : \quad f_{tl,p} \geq 0 \tag{35}$$

$$\forall tl \in TL, \forall r \in R(tl), \forall B \in B'_r : \quad \lambda^B \geq 0 \tag{36}$$

$$\forall tl \in TL, \forall p \in P : \quad y_{tl,p} \geq 0 \tag{37}$$

To determine which transport resource packings to add to the set B'_r , we have to check the reduced cost of each transport resource packing we have not yet added to the set. The reduced cost of a transport resource packing B called $\bar{c}_B$ is given by the following equation. We then could add the transport resource packing with the lowest reduced cost to the set B'_r . For all $tl \in TL, r \in R(tl)$ and $B \in B_r \setminus B'_r$ the reduced cost of the transport resource packing λ^B is given by:

$$\begin{aligned}
\bar{c}_B = & cost(tr(r)) * distance(r) + co2emissions(tr(r)) * distance(r) * co2costs \\
& + \sum_{p \in P} \alpha_p^B * compoundInterest(value(p), distance(r)/speed(tr(r))) \\
& - \sum_{p \in P} \alpha_p^B * \pi_{tl,p}
\end{aligned} \tag{38}$$

4.4 Pricing Problem

However, instead of enumerating all possible transport resource packings, we can use an auxillary optimization problem called the pricing problem to determine which transport resource packings

to add to the set B'_r . The objective function of the pricing problem is given by the equation (39) and we perform a pricing problem for each transport link $tl \in TL$ and each route $r \in R(tl)$:

$$\begin{aligned} \min \quad & cost(tr(r)) * distance(r) + co2emissions(tr(r)) * distance(r) * co2costs \\ & + \sum_{p \in P} \mathbf{x}_p^r * compoundInterest(value(p), distance(r)/speed(tr(r))) \\ & - \sum_{p \in P} \mathbf{x}_p^r * \pi_{tl,p} \end{aligned} \quad (39)$$

where for every product p which can be transported on the route r with the transport resource $tr(r)$ associated with the route r , we introduce an integer variable $\mathbf{x}_p^r \in \mathbb{Z}_+$ which represents how many units of product p are packed into the packing.

Because it is unlikely that many different transport resources are used between two locations, we only add the transport resource packing out of all routes $r \in R(tl)$ on each transport link $tl \in TL$ which has the minimal reduced cost. Doing this prevents from adding many transport resource packings which most likely will not be used in an optimal solution. Therefore, we only add at most one transport resource packing for each transport link $tl \in TL$ to the set B'_r at each iteration of the column generation algorithm.

In a valid transport resource packing B the total size of the products packed into the transport resource packing must not exceed the capacity of the transport resource used on the route. This constraint is given by the following equation:

$$\sum_{p \in P} size(p) * \mathbf{x}_p^r \leq capacity(tr(r)) \quad (40)$$

To estimate the performance of our linear program formulation, we can have a look on the number of constraints and pricing problems:

$$numberOfConstraints = |L| * |P| + |TL| * |P| \quad (41)$$

$$= (|L| + |TL|) * |P| \quad (42)$$

$$numberOfPricingProblems = \sum_{tl \in TL} |R(tl)| \quad (43)$$

5 Experimental Results

To evaluate the performance of our algorithm, we have implemented the proposed Master Problem and Pricing Problem in C++ using the SCIP Optimization Suite by K. Bestuzheva et al. [5]. The experiments were run on a 64 bit Windows 10 machine with an AMD Ryzen 5 PRO 5650U CPU and 16 GB of RAM. We used the precompiled SCIP Optimization Suite 8.0.4 and the Soplex 6.0.4 solver was used to solve the linear and integer programs. All reported times are the solving times outputted by SCIP.

We have tested our algorithm on the example instances given in 1.2. The transport network of the examples depicted in figure 1 contains 6 locations, 10 transport links, 19 routes and 2 transport resources. Therefore all example instances have 19 pricing problems.

The first example depicted in figure 2 contains only one product. Therefore the resulting Master Problem has $(6 + 10) * 1 = 16$ constraints. It was solved in 0.177 seconds and in total 8 packings were added by the pricing problems.

The second example depicted in figure 3 contains two products. Therefore the resulting Master Problem has $(6 + 10) * 2 = 32$ constraints. It was solved in 0.318 seconds and in total 22 packings were added by the pricing problems.

To evaluate the performance of our algorithm on larger instances, we got two example instance from Airbus. The first instance contains 17 locations, 216 transport links, 906 routes, 16 transport resources and 20 products. Therefore the resulting Master Problem has $(17+216)*20 = 4660$ constraints. It was solved in 9.958 seconds and in total 93 packings were added by the pricing problems.

The second instance from Airbus contains 50 locations, 1532 transport links, 5610 routes, 16 transport resources and 35 products. Therefore the resulting Master Problem has $(50 + 1532) * 35 = 55370$ constraints. It was solved in 305. seconds and in total 4474 packings were added by the pricing problems.

6 Outlook

The main limitation of our approach is that it produces solutions which use some transport resource packings only very infrequent. As an example we consider a product which is quite small in size and has to be shipped from $a \in L$ to $b \in L$ at a rate of 20 per month. Therefore, it is possible to pack many units of this product into a transport resource with a high capacity e.g. a Beluga. If the product fits for example 500 times into a Beluga this packing which only includes this one product will be used at a frequency of 0.04 per month or 0.48 per year. However, this is not feasible in the real world as high storage costs would occur. Therefore, a future improvement could be to take into account the storage costs of the products.

Additionally we do not take into account disruptions that could occur in the network. In general one could add a delay distribution to each transport route and take into account the probability of a delay occurring.

A further improvement could be to take into account the lead time of a produced root product, modelling a bill of materials which results in a tree structure to model the production of the root product. By taking into account the production time of the products we could model the problem more accurately and take into account the time it takes to produce and transport a product. The resulting lead time of the product could then be used improve the real world performance of the network.

7 Conclusion

In this thesis we have presented a linear program formulation for the problem of finding the optimal transport network for a given set of products. We have shown that the linear program formulation is equivalent to the problem described in Section 1.1. Additionally, we have presented a column generation approach to solve the linear program. We have evaluated the performance of our algorithm on the example instances given in Section 1.2 and we have discussed the limitations of our approach and pointed out possibilities for future improvements.

In conclusion, linear program formulation presented offers a simple and effective way to model the problem. Additionally, the column generation approach allows us to solve the linear program in a reasonable amount of time. Through this formulation decision makers can easily evaluate the impact of adding new products to the network or changing the production and consumption rates of the products. Therefore, the linear program formulation can be used to support decision makers in the process of designing a transport network.

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Declaration of originality

I hereby declare in lieu of an oath that I have written this thesis independently and have not used any resources other than those specified - in particular no Internet sources not named in the list of sources. The parts of the work that are taken from other works (this also includes internet sources) in wording or in spirit have been identified and the source has been indicated. I further assure that I have not previously submitted the work in another examination procedure and that the written version submitted corresponds to that on the electronic storage medium.

Karl Henning